Proof : Take $C \neq C'$

outer code will differ in $\geq d_{out}$ locations

Each such location, when fed into inner code, o/p will differ

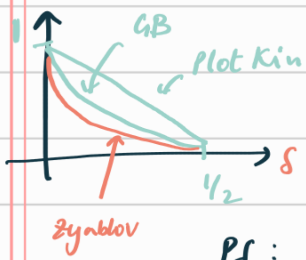
$\geq d_{in}$ locations.

RS code as outer code. $\circ$ Asymp. Good Binary Code as inner code $=$ Asymp. Good Binary Code

Zyablov Bound

(Binary Codes)

For any $\varepsilon > 0$, there exists a family of codes which are binary & Explicit, with a symbol rate R & distance δ.



$$R \geq \sup_{r \in (0, 1-H(\delta)-\varepsilon)} r\left(1 - \frac{\delta}{H^{-1}(1-r-\varepsilon)}\right)$$

Pf : Via concatenate code

$C_{out} \to$ RS code with rate R_{out}, $\delta_{out} = 1 - R_{out}$

$C_{in} \to$ Binary l.h code which lies on the GV curve $r \geq 1 - H(\delta_{in}) - \varepsilon$

• Larger r will make δ_{in} small.

we dont want δ_{in} to be smaller than δ. Hence r has an upper bound.

$$\delta \geq \delta_{in}\,\delta_{out} \geq (1 - R_{out})\, H^{-1}(1-r-\varepsilon)$$

$$R = \underbrace{R_{in}}_{r} \cdot R_{out} \geq r \cdot \left(1 - \frac{\delta}{H^{-1}(1-r-\varepsilon)}\right)$$

RS code: $q = 2^{k_{in}}$

Eval pts $\to$ $\mathbb{F}_q^* = q - 1 = n_{out} = 2^{k_{in}} - 1$

$$k_{in} = \log q \approx \log(n_{out})$$

$$n_{in} = \frac{k_{in}}{r} = \frac{\log(n_{out})}{r} \leq \frac{\log n}{r}$$

Go over all possible $k_{in} \times n_{in}$ generator matrices

$$2^{k_{in} \times n_{in}} \leq 2^{\log^2 n / r} \leq n^{O(\log n)}$$

Enumerate all possible binary vectors of length $n_{in}-K_{in}$

complexity in each phase $\to$ $2^{n_{in}-K_{in}} \leq 2^{O(\log n)} \leq n^c$

Justesen Code

① Dont have to use same inner code $\forall$ outer code symbols

② Suffices to show that atleast some $(1-\varepsilon)$ fraction of inner code is good.

Outer code is R.S. $(\mathbb{F}_q^*, n_{out}, K_{out})$ $\quad R_{out} = \frac{K_{out}}{n_{out}}$

$q = 2^{K_{in}}$

$n_{out} = q-1$

Each element of outer code : $(\omega_1, \omega_2, \cdots, \omega_{n_{out}})$

will map with rate $= 1/2$ {of inner code}

$n_{in} = 2K_{in}$

$$C = \left\{ C_{in}^1(f(\alpha_1)) \cdot C_{in}^2(f(\alpha_2)) \cdots\cdots C_{in}^{n_{out}}(f(\alpha_{n_{out}})) : f \in \mathbb{F}_q[x],\ \deg(f) \leq K_{out} \right\}$$

Assume that $(1-\varepsilon)$ fraction of inner codes satisfy GV bound

ε very small $\qquad \delta_i \geq H^{-1}\left(\frac{1}{2}-\varepsilon\right)$

Proposition: For $R_{out} \in (0,1)$; Justesen code is asymptotically good with rate $= \frac{R_{out}}{2}$.

Proof: Need to show (+ve) minimum distance in limit.

Min weight of a non zero codeword

Outer code: atleast $\frac{n_{out} - K_{out} + 1}{n_{out}}$ fraction of outer symbols will be non zero

$\approx 1 - R_{out}$

Choose ε s.t. $1 - R_{out} > 2\varepsilon > 0$

$\rightsquigarrow$ Atmost ε of inner are bad

$\rightarrow$ Atleast ε fraction will non zero symbols which will be mapped via good codes.

$\Rightarrow \quad d \geq (\varepsilon \cdot n_{out}) \cdot (2K_{in} \cdot \delta_i)$ $\qquad \delta_i \geq H^{-1}\left(\frac{1}{2}-\varepsilon\right)$

Lower bound on relative distance $\delta \geq \varepsilon \cdot H^{-1}(1/2 - \varepsilon) > 0$

$\delta \geq H^{-1}(1-R-\varepsilon)$

Wozencroft Ensemble

From GV, $\exists$ $\left(R \geq 1 - H(\delta) - \varepsilon \rightarrow H(\delta) \geq 1 - R - \varepsilon\right)$

Claim: $\varepsilon > 0$, k large enough $\exists$ a family of $(2k, k)$ codes

$C_{in}^{1}, C_{in}^{i} \dots C_{in}^{N}$ $\quad N = 2^k - 1$

s.t. for $\geq (1-\varepsilon)N$ of these C_{in} has a rel. dist. of $\geq H^{-1}(\frac{1}{2} - \varepsilon)$

Outer code - $RS(\mathbb{F}_q^{*}, n_{out}, k_{out})$

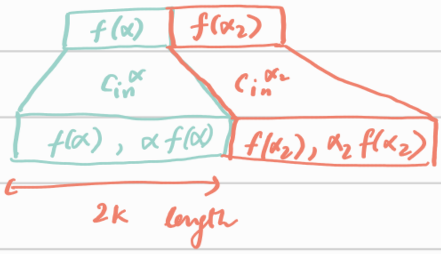
$\mathbb{F}_q$ $\quad N = q - 1, \; q = 2^k$

$(= n_{out})$

$C_{in}^{\alpha} : \mathbb{F}_2^{k} \longrightarrow \mathbb{F}_2^{2k}$

$C_{in}^{\alpha}(x) = (x, \alpha x)$

RS codeword each symbol is an evaluation $f(\alpha)$, $\alpha \in \mathbb{F}_q^{*}$



Claim: $C_{in}^{\alpha_1}, C_{in}^{\alpha_2}$ for $\alpha_1 \neq \alpha_2$, do not share any codeword.

Pf: Assume $y \in C_{in}^{\alpha_1}$ & $y \in C_{in}^{\alpha_2}$

$\Rightarrow \quad y = (y_1, \alpha_1 y_1)$

$\quad\quad y = (y_2, \alpha_2 y_2)$ $\quad \Rightarrow \alpha_1 = \alpha_2 \Rightarrow$ contradiction

$\{$ # bad of codes in this family is bounded by

\# 2k-length vector with HW $\leq 2k \cdot H^{-1}(1/2 - \varepsilon)$

As no common code word, the no. of vectors with HW $\leq 2k \cdot H^{-1}(1/2 - \varepsilon)$ to be each can be assumed in a different family.

$y \in \mathbb{F}_2^{2k}$, $wt(y) \leq 2k \cdot H^{-1}(1/2 - \varepsilon)$

$\leq Vol\left(2k, H^{-1}(\frac{1}{2} - \varepsilon) 2k\right)$

$\leq 2^{2k \cdot H(H^{-1}(1/2 - \varepsilon))} = \underbrace{2^{k - 2k\varepsilon} \leq \varepsilon(2^k - 1)}_{\text{For large enough } k} = \varepsilon N$

Inner Code family

$R \in (0,1)$

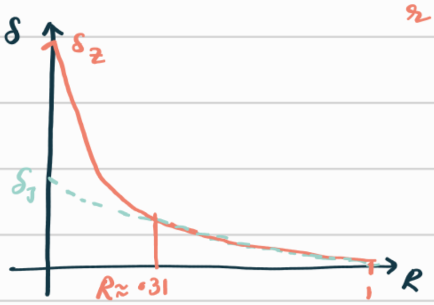
$r \in (1/2, 1)$

$C_{in}^{\alpha} = (x, (\alpha x)_\ell) = \frac{k}{k+\ell}$

Justesen code: for $R \in (0,1)$ $\exists$ explicit binary linear code of rate $\geq R$

& $\delta \geq \max_{r \geq \max(1/2, R)} \left(1 - \frac{R}{r}\right) H^{-1}(1-r-\varepsilon)$

Additional constraint, which allows for EXPLICIT CONSTRUCTION.

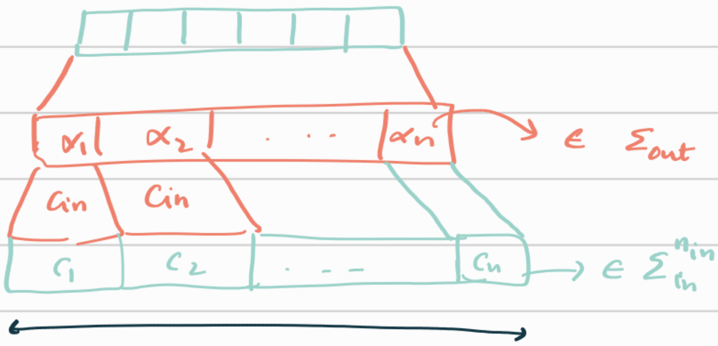


Zyablov: $\delta_Z \geq \max_{r \geq R} \left(1 - \frac{R}{r}\right) H^{-1}(1-r-\varepsilon)$

Decoding of concatenated codes

$x \in \Sigma_{in}^{k_{in} \times k_{out}} = \Sigma_{out}^{k_{out}}$

$e < \frac{d_{in} d_{out}}{2} - 1$



$y_i \to$ the i^{th} recieved word

I^{st} Attempt: ① Decode each of the inner code blocks: $\hat{c}_i$ closest to y_i $\in C_{in}$

② Use C_{in}^{-1} to convert $\hat{c}_i$ to $\hat{\alpha}_i \in \Sigma_{out}$

③ Decode C_{out} to get original message

Call an output block bad if # errors is $\geq \left\lfloor \frac{d_{in}-1}{2} \right\rfloor$

Total no of errors is e then

$\#$ bad blocks $\leq \frac{e}{\left\lfloor \frac{d_{in}-1}{2} \right\rfloor}$

Decode correctly if $\frac{e}{\left\lfloor \frac{d_{in}-1}{2} \right\rfloor} \leq \left\lfloor \frac{d_{out}-1}{2} \right\rfloor$

$\Rightarrow e \leq \left\lfloor \frac{d_{out}-1}{2} \right\rfloor \left\lfloor \frac{d_{in}-1}{2} \right\rfloor$

Developing intuition for the good decoding strategies:

Error pattern: ① Add d_{in} errors to as many blocks as possible

$\frac{e}{d_{in}}$ blocks which are erroneous.